

Let R be the relation that contains the pair (a, b) if a and b are cities such that there is a direct nonstop airline flight from a to b . When is (a, b) in

- a) R^2 ? b) R^3 ? c) R^* ?

Suppose that the relation R is reflexive. Show that R^* is reflexive.

Use Warshall's algorithm to find the transitive closures of the relations in Exercise 26.

- a) $\{(a, c), (b, d), (c, a), (d, b), (e, d)\}$
b) $\{(b, c), (b, e), (c, e), (d, a), (e, b), (e, c)\}$
c) $\{(a, b), (a, c), (a, e), (b, a), (b, c), (c, a), (c, b), (d, a), (e, d)\}$
d) $\{(a, e), (b, a), (b, d), (c, d), (d, a), (d, c), (e, a), (e, b), (e, c), (e, e)\}$